

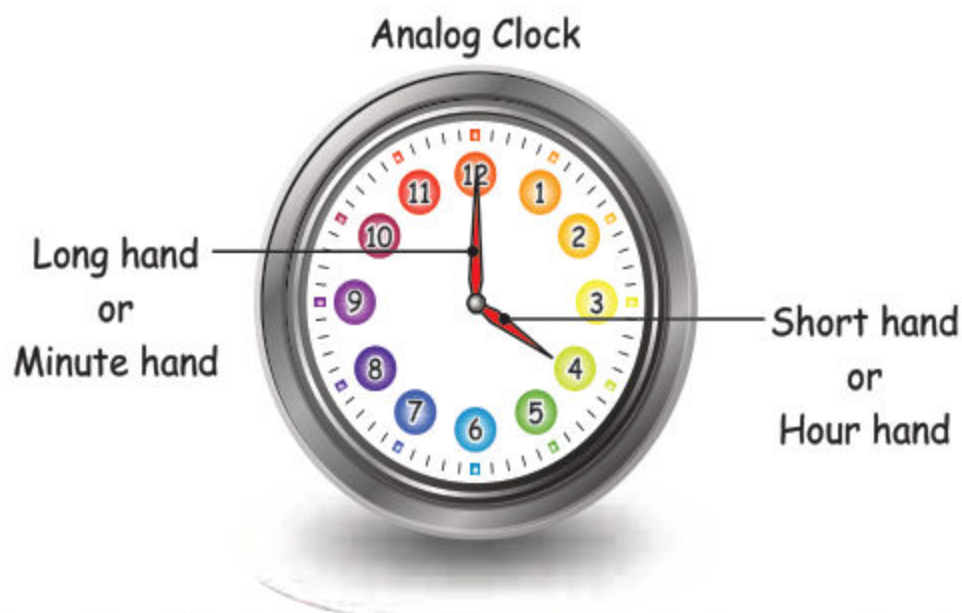
In Grade 1, we learnt to read and show time in hours (o'clock). In this unit, we will learn to read and show time in hours and minutes. Additionally, we will recognize the duration of time to estimate a rough calculation of the time taken by specific events or tasks.

Time in hours and minutes on an analogue clock

Do you know how to tell time on a clock with hands? We use analogue clocks to show time in hours and minutes. Let us first know about the analog clock.

What is analog clock?

An analog clock is a type of clock that displays time using hour and minute hands, also known as clock hands or pointers. It is a traditional clock face with a circular dial, marked with numbers from 1 to 12, and two hands:



1. Short hour hand (or hour pointer): points to the hour on the clock face.
2. Long minute hand (or minute pointer): points to the minutes on the clock face. The long hand (minute hand) tells minutes by pointing to the numbers on the clock face. Each number on the clock face represents 5 minutes, and the long hand moves in small increments (usually 1-5 minutes) around the clock face.

Example 1:

Tell the time if short hand is on 7 and long hand is on 3.

As the short hand is on the 7, it means 7 hours have passed. Meanwhile, the long hand points to the 3, which means 15 minutes ($3 \times 5 = 15$). Thus, the clock shows 7 hours and 15 minutes and we write it as 7:15.



Example 2:

Tell the time if short hand is on 5 and long hand is on 8.

As the short hand is on the 5, it means 5 hours have passed. Meanwhile, the long hand points to the 8, which means 40 minutes ($8 \times 5 = 40$). Thus, the clock shows 5 hours and 40 minutes and we write it as 5:40.



Example 3:

Tell the time if short hand is on 6 and long hand is on 7.

As the short hand is on the 6, it means 6 hours have passed. Meanwhile, the long hand points to the 7, which means 35 minutes ($7 \times 5 = 35$). Thus, the clock shows 6 hours and 35 minutes and we write as 6:35.



Exercise 1

Tell the time



Interval (Duration) of Time

What is a Time Interval?

- A time interval is the amount of time it takes to do something.
- It is like measuring how long something lasts.

Example 1:

A boy eats biryani at 2:00 pm and finishes it on 2:15 pm. How long does he take to finish it.

Solution:

He takes 15 minutes to finish the Biryani.



Example 2:

If boys start playing football on 5:00 pm and end the game on 7:00 pm. How long does it take to play the football match?

Solution:

Here, the boys take 2 hours to play a football match.



Example 3:

If Ali starts watching movie on 8:00 pm and it lasts till 11:00 pm.

How long does it take to watch a movie?

Solution:

Here, it takes 3 hours to watch a movie.





Activity

- (i) How long do you watch a cartoon?
- (ii) How long do you take sleep?
- (iii) How long do you study at school?
- (iv) How long do you take to visit an amusement park?

Exercise 2

- (1) If Khizar starts watching cartoon on 6:00 pm and finishes it on 6:30 pm. How long does it last?

Solution:

He takes _____ minutes to watch the cartoon.

- (2) If Asra goes on short trip with her parents from 10:00 am to 1:00 pm. How long has she been on short trip?

Solution:

She has been on short trip for the _____ hours.

- (3) If an artist draws a picture from 3:00 pm to 5:30 pm. How long does he draw the picture.

Solution:

He takes _____ to draw a picture.

- (4) If Hina leaves Karachi at 9:00 am and arrives Hyderabad at 11:00 pm. How long does she travel?

Solution:

She travels _____ from Karachi to Hyderabad.

Use of Calendar

In our previous class, we learnt the names of the days of the week and months in a calendar. Now, we will use this knowledge to find a particular day or date within a week or month using the calendar.

Names of days in a week



Monday

1st day



Tuesday

2nd day



Wednesday

3rd day



Thursday

4th day



Friday

5th day



Saturday

6th day



Sunday

7th day



Activity

- (i) What day is today?
- (ii) What day is tomorrow?
- (iii) What day was yesterday?

Names of months in a year:



Activity

- (i) What is the current month?
- (ii) What will be the next month?
- (iii) What was the previous month?

Unit 7: Time

Modal Solar Calendar:



We solve following examples using above modal solar Cylinder.

Example 1:

Rida's birthday is on 13th October. Which day of the week will it be?

Solution:

We are only focusing on the month of October to find out which day Rida's birthday is.



Thus, the calendar shows that Rida's birthday is on Monday.

Example 2:

The school's annual award ceremony day is scheduled for the 2nd of April. Which day of the week will it be?

Solution:

We are only focusing on the month of April to find out on which date school's annual award ceremony day is.



Thus, the calendar shows that school's annual award ceremony day is scheduled on Wednesday.

Example 3:

Family of Sarfraz plans a picnic on the first Sunday of January. What date will their picnic be?

Solution:

We are only focusing on the month of January to find out on which day Sarfraz family goes to picnic.



Thus, Sarfraz family goes to the picnic on 5th of January.

Example 4:

Grandmother is coming to visit on the 3rd Thursday of July. What date will it be?

Solution:

We are only focusing on the month of July to find out on which date Grandma is coming to visit.



Thus, grandmother is coming on 17th of July to visit.

Exercise 3

1. Ahmad's birthday is on 28th June. What day of the week will it be?

Solution:

The day of the Ahmad's birthday is _____.

2. The school holiday is on 11th November. What day of the week is it?

Solution:

The day of the School holiday is _____.

3. Kamran's family plans a picnic on the second Saturday of June. What date will their next picnic be?

Solution:

The date will be _____.

4. Sport festival of the school is conducted on third Monday of February. What date will it be?

Solution:

The date will be _____.

5. On what day of week, the Independence Day of our country is celebrated?

Solution:

The day of the independence is _____.

GEOMETRY

Student Learning Outcomes:

At the end of the chapters, Students will be able to:

- ✓ Recognize, identify and draw 2-D shapes (Rectangle, square, circle, triangle, semi-circle and quarter-circle) with respect to their characteristics.
- ✓ Recognize and identify 3-D shapes (cube, cuboid, cone, cylinder and sphere) in different orientations.
- ✓ Describe the position, direction and movement of an object including moving clockwise, anti-clockwise, quarter, half and three quarters turns using appropriate positional language (for instance: inside, outside, above, below, over, under, far, near, before, after, besides, between, left, right and in front of, quarter turn, half turn, three quarter turns, clockwise, anti-clockwise, behind etc.)

